

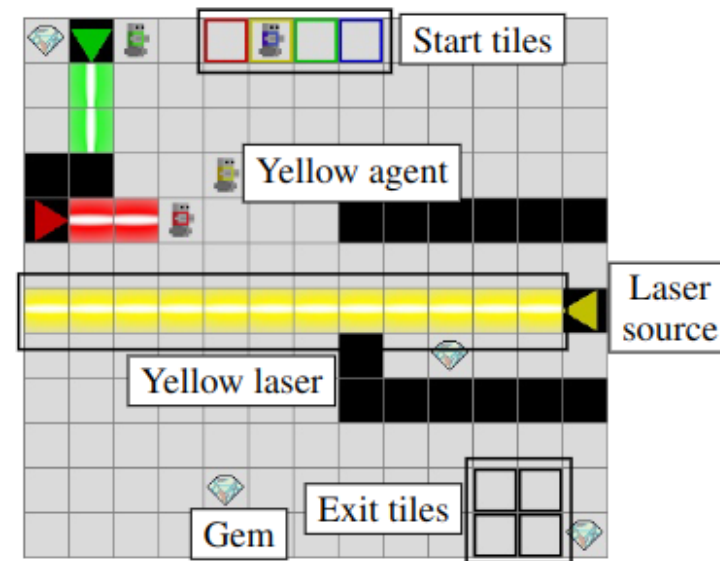


Procedural Generation of Solvable Cooperative Levels for the Laser Learning Environment

Hugo Charels — Master thesis defense

Supervisors: Tom Lenaerts · Yannick Molinghen

ULB — Academic year 2025-2026 · 18/06/2026



Context & motivation

Laser Learning Environment (LLE)

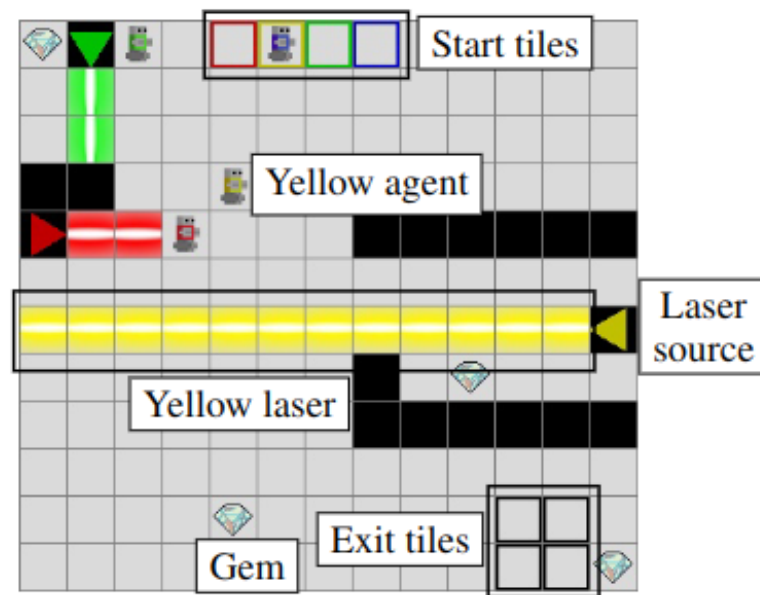
- 2D grid-based fully cooperative multi-agent puzzle
- Each agent must reach its exit tile **simultaneously** with all others
- Laser beams are passable only by an agent of the **matching color**
- Reward signal is sparse: zero credit for intermediate coordination steps

The LLE cooperative mechanic

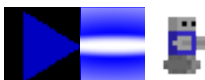
- A beam blocks **other** colours; each agent is **immune** to its own.
- Block your own beam → free a teammate's path (**locally unrewarded**).

Objective

Generate LLE levels **solvable** and that **require cooperation**.



LLE Level 6 — the hard target



Part 1 – Verification (RQ1–2)

How can we formally verify that a level is solvable and that solving it genuinely requires cooperation?

In this part we present our SAT-based verification tool:

RQ1 – Bounded-horizon solvability

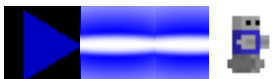
- Formalise the decision problem
- Reduce to SAT: level + horizon $\rightarrow$ Boolean formula
- Verify solvability with a SAT solver

RQ2 – Cooperation detection

- Reuse the SAT encoding, change one rule
- Detect whether cooperation is required
- Classify cooperation structures

Key idea

SAT solver as a verification oracle: given a level and a horizon $T_{\max}$, answer “solvable?” in bounded time.



Bounded-horizon solvability $\rightarrow$ SAT

Decision problem

Given a level L and a horizon $T_{\max}$, does a valid joint trajectory exist?

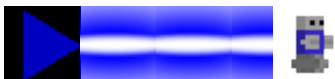
SAT reduction

Build a Boolean formula $\Phi(L, T_{\max})$. Satisfiable $\Leftrightarrow$ solvable within $T_{\max}$.

$(L, T_{\max}) \rightarrow$ encode Φ $\rightarrow$ SAT solver $\rightarrow$ SAT / UNSAT $\rightarrow$ solvable / unsolvable

Why SAT? Fast, well-known problem, modern solvers are highly efficient.

Complexity: LLE solvability $\in$ NP and $\leq_p$ SAT. NP-hardness **open**.



Detecting cooperation

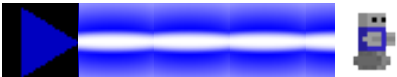
Idea: keep the SAT encoding, change **one** rule: beams no longer block same-colour agents → the cooperation mechanic is **disabled**.

Cooperation criterion

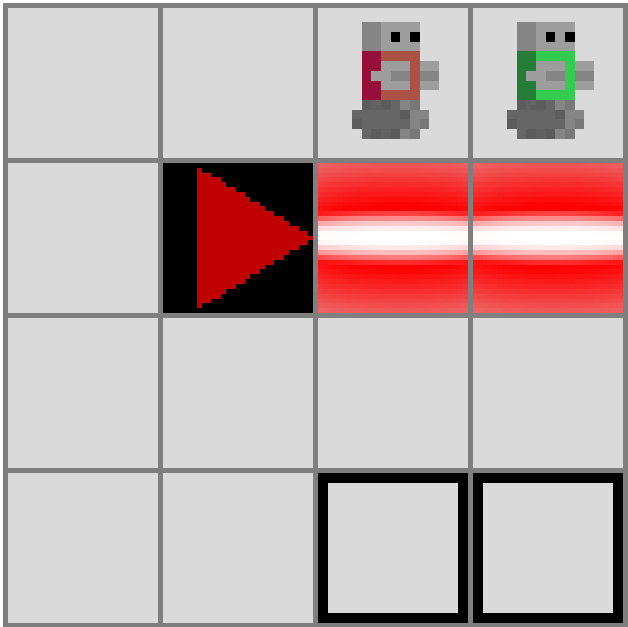
L requires cooperation with $T_{\max}$ $\iff \Phi(L, T_{\max})$ **SAT** and $\Phi_{\text{strict}}(L, T_{\max})$ **UNSAT**.

Three possible outcomes:

Condition	Label	Meaning
Φ UNSAT	UNSOLVABLE	No valid trajectory exists
Φ_{strict} UNSAT	COOPERATIVE	Every solution blocks a beam – cooperation required
Otherwise	NON_COOPERATIVE	A solution exists without the blocking mechanic



Horizon matters



Horizon	Solvable	Coop.
$T_{\max} \leq 2$	no	n/a
$3 \leq T_{\max} \leq 8$	yes	yes
$T_{\max} \geq 9$	yes	no

- **Too small** → agents run out of moves; the level is falsely marked unsolvable
- **Too large** → agents can take longer detours that bypass cooperation requirements

Trade-off

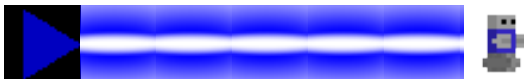
Larger $T_{\max}$ → more actions (more solutions), but bigger formula (longer solve duration). Pick the **smallest** $T_{\max}$ that still admits a solution.



Cooperation profiles

From one SAT model $\rightarrow$ a **dependency graph** (edges: helper $\rightarrow$ beneficiary)

classify by priority: fully coupled $\succ$ mutual $\succ$ distributed $\succ$ chain $\succ$ asymmetric.

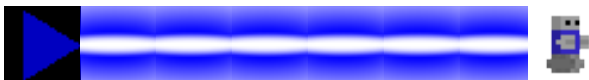
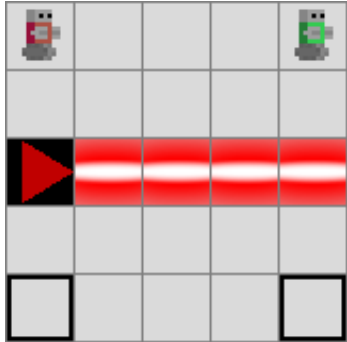


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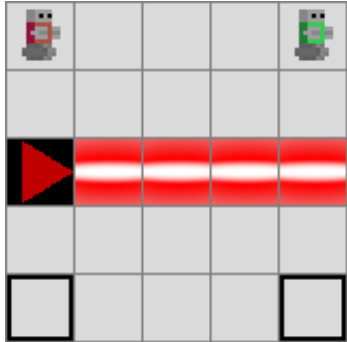


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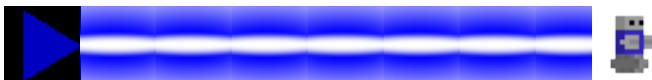
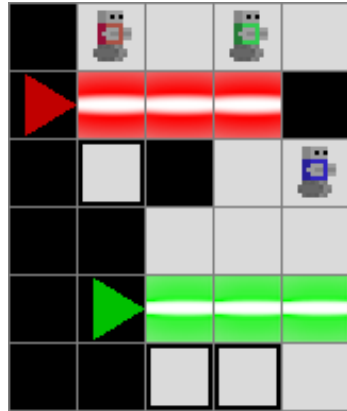
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asymmetric



chain

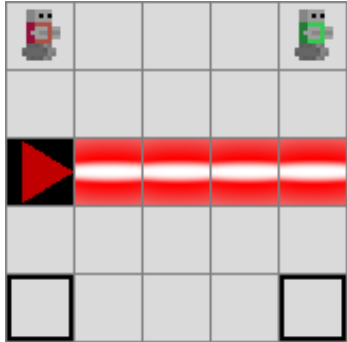


Cooperation profiles

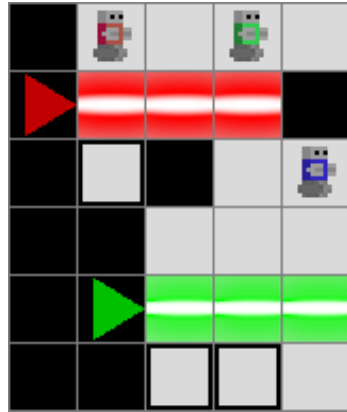
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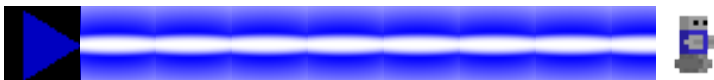
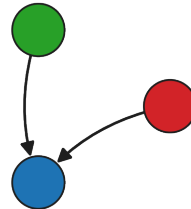
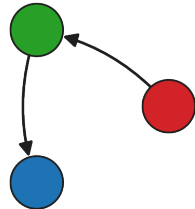
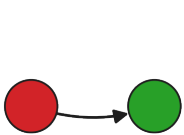
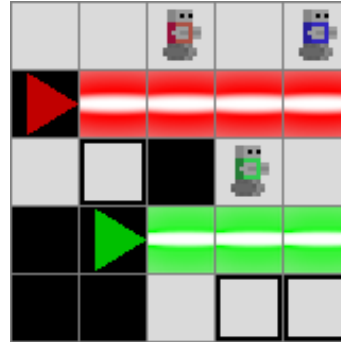
asymmetric



chain



distributed

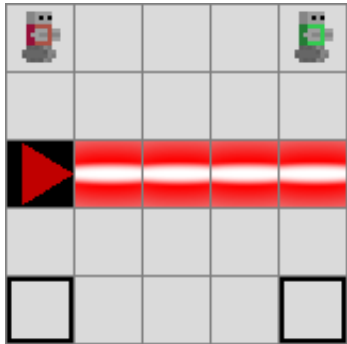


Cooperation profiles

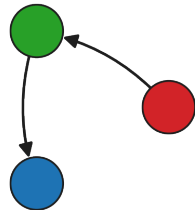
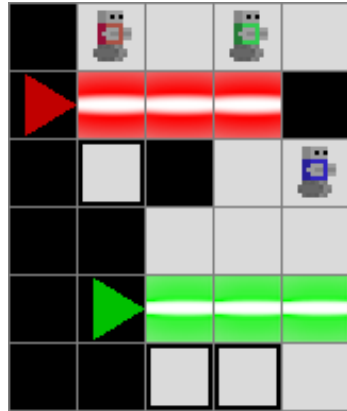
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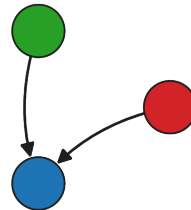
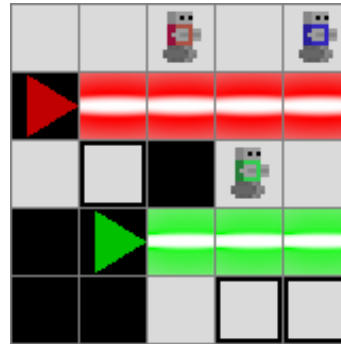
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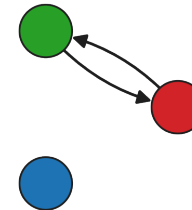
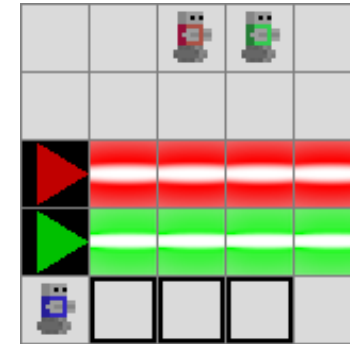
chain



distributed



mutual

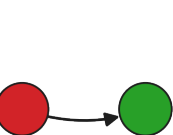
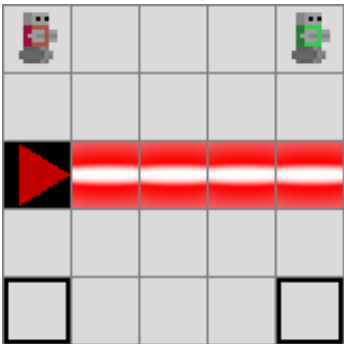


Cooperation profiles

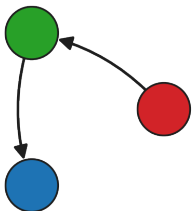
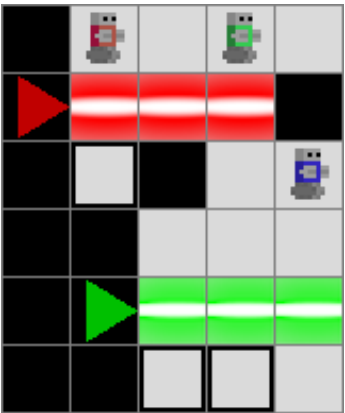
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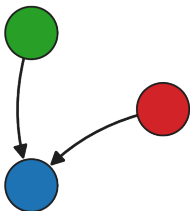
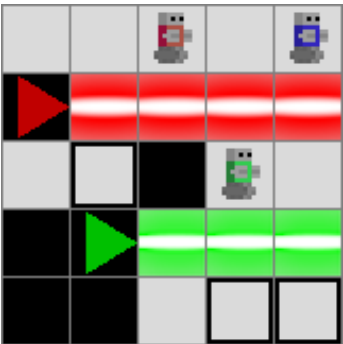
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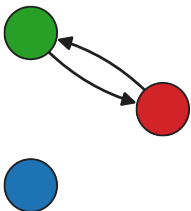
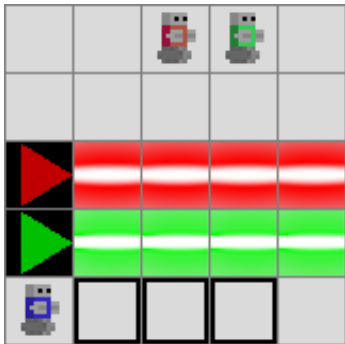
chain



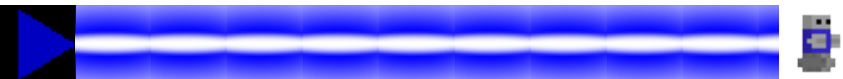
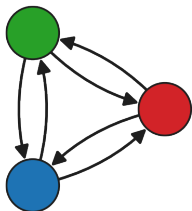
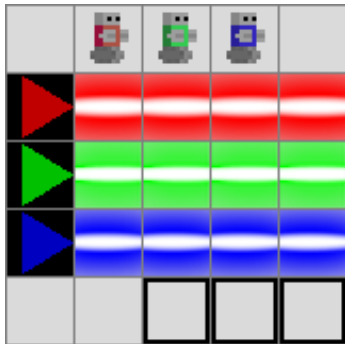
distributed



mutual



fully coupled



Part 2 – Construction (RQ3–4)

How can we embed formal verification inside a procedural generator so that every accepted level is certified, and can we control the cooperation structure by targeting specific profiles?

In this part we present our level generators:

RQ3 – Generator efficiency

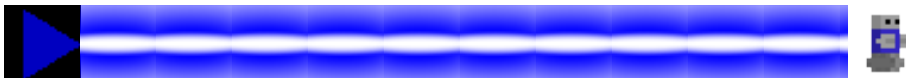
- Random placement with geometric constraints
- Constructive lane-based approach
- Level-6-style clustered design
- SAT oracle filters solvable / cooperative levels

RQ4 – Output diversity

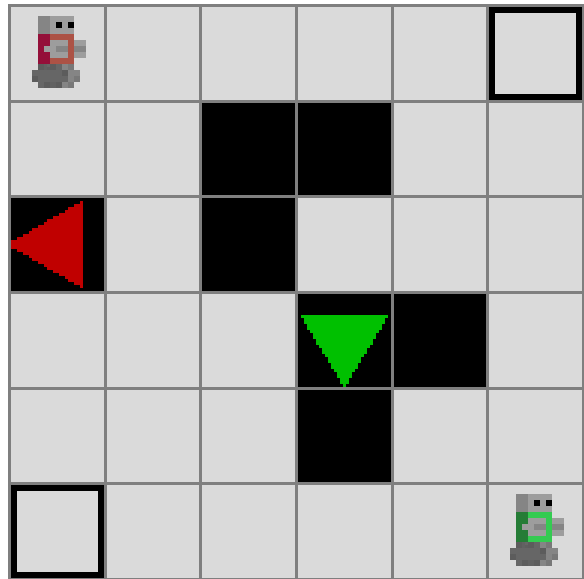
- Five-profile taxonomy of cooperation structures
- Generators can steer towards specific profiles
- Constructive → mutual by design

Key idea

Place-and-check with a SAT oracle: every accepted level is **provably** solvable.

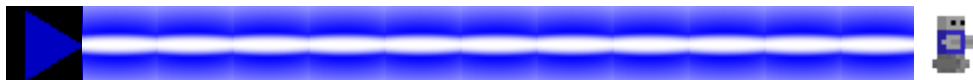


Random Generator

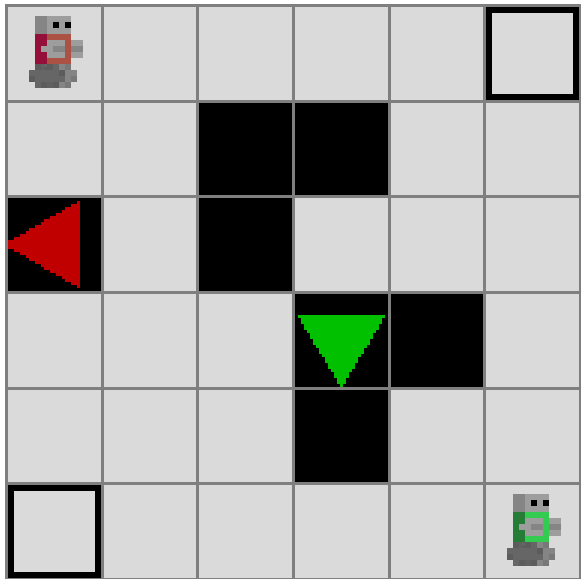


Pure random

Lasers point off-grid, a wall block a laser source

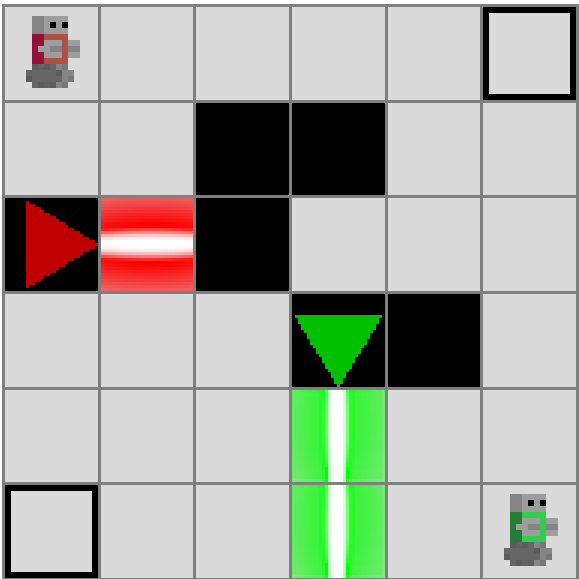


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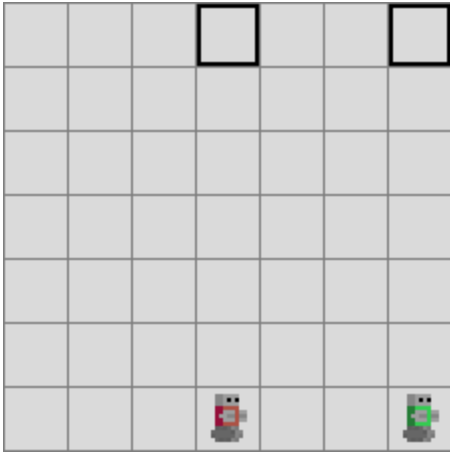
Constrained random

Filters out laser sources that behave like walls.

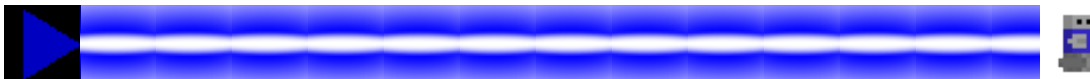


Constructive Generator

Oriented **lane-based** design. Each agent gets a reserved lane, cooperation arises from crossing lasers. Near-zero rejection rate.

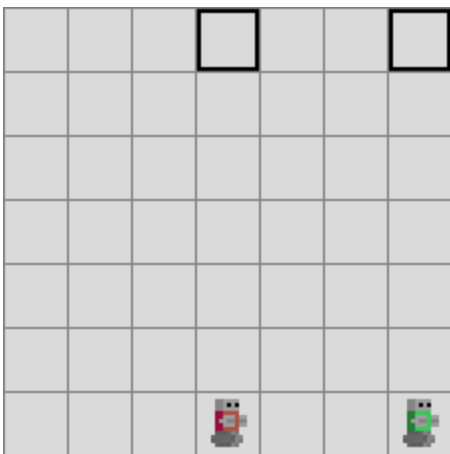


1. Reserve lanes, place agents + exits

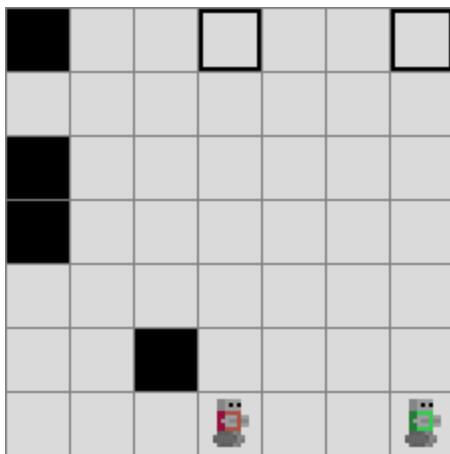


Constructive Generator

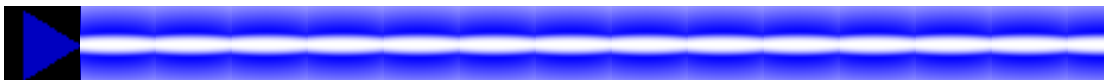
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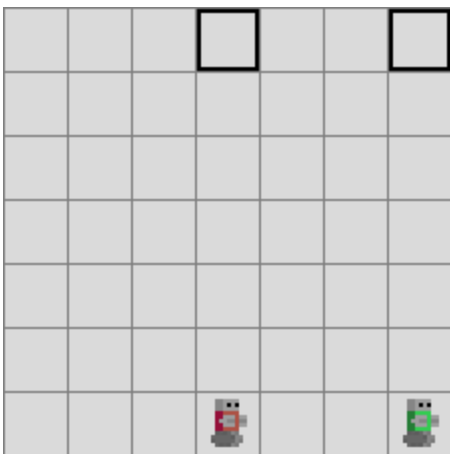


2. Place walls in free cells

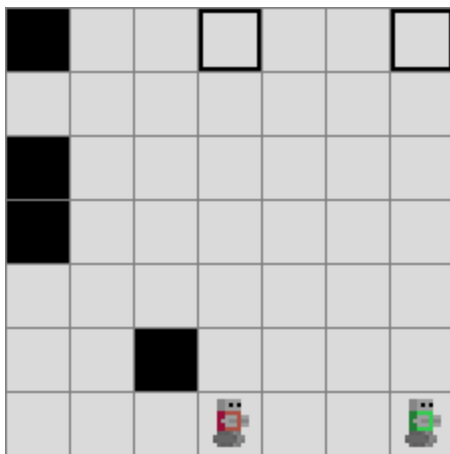


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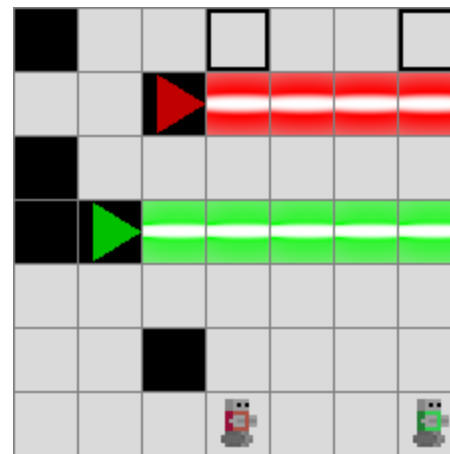
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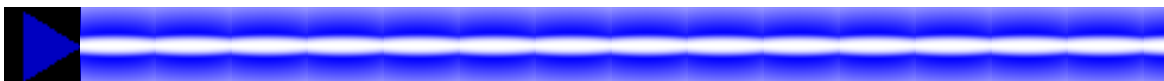
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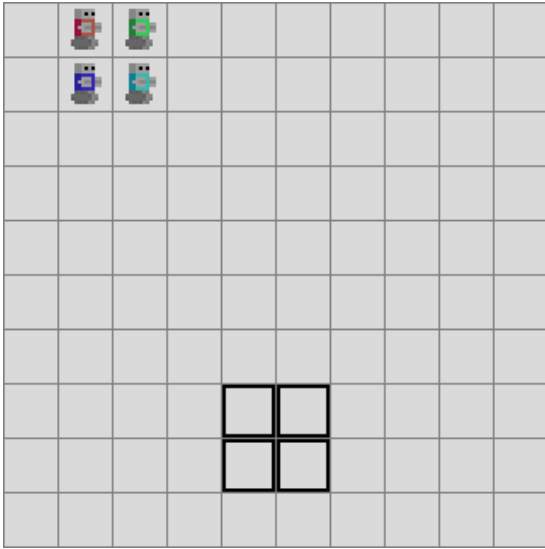


3. Add lasers crossing lanes

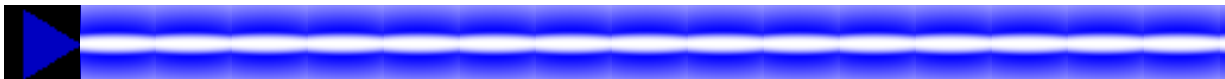


Level-6-Style Generator

Clustered starts + exits on opposite sides. Shared laser corridor = mutual by design. SAT verifies.

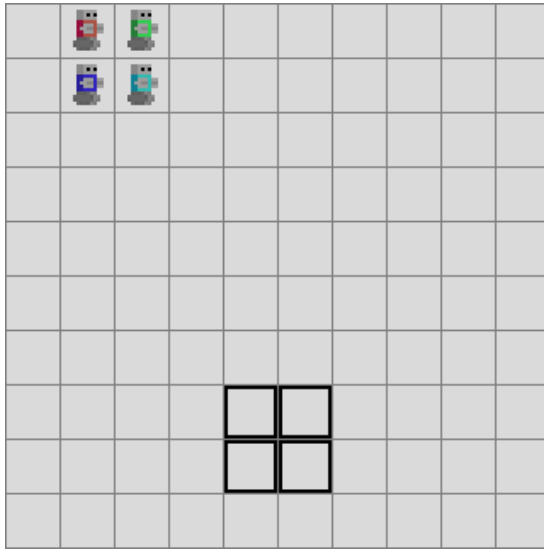


1. Place clusters on opposite sides

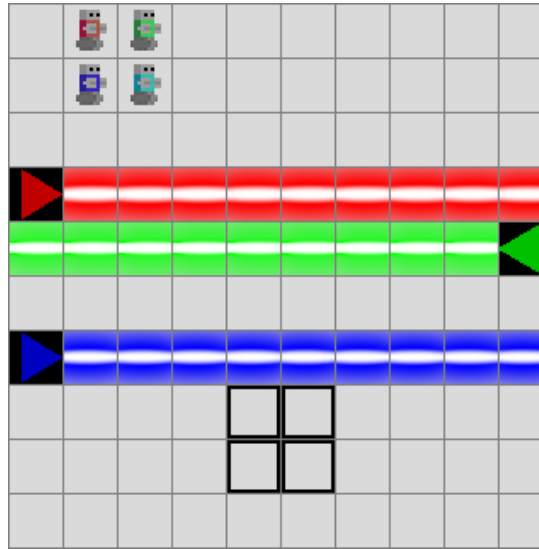


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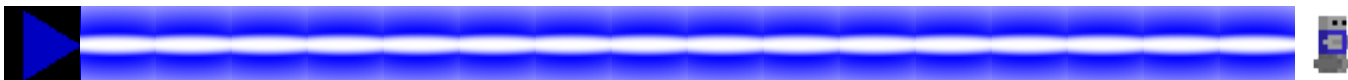
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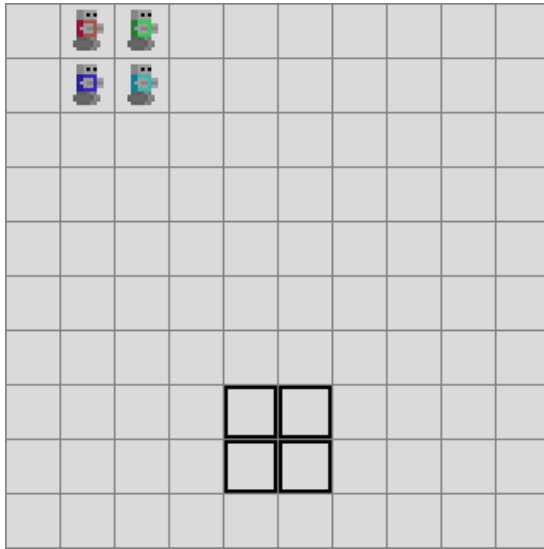


2. Fill the central corridor with lasers

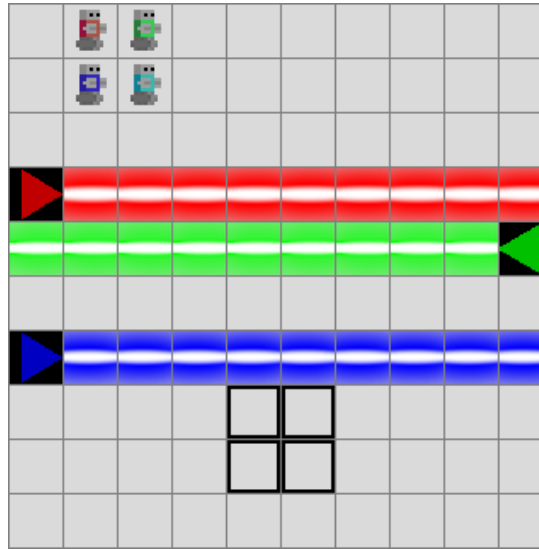


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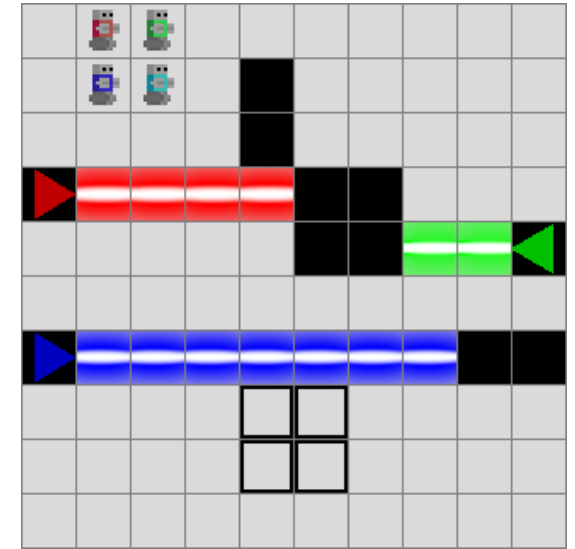
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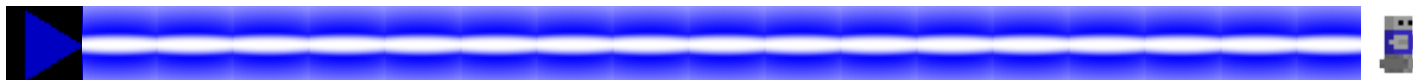
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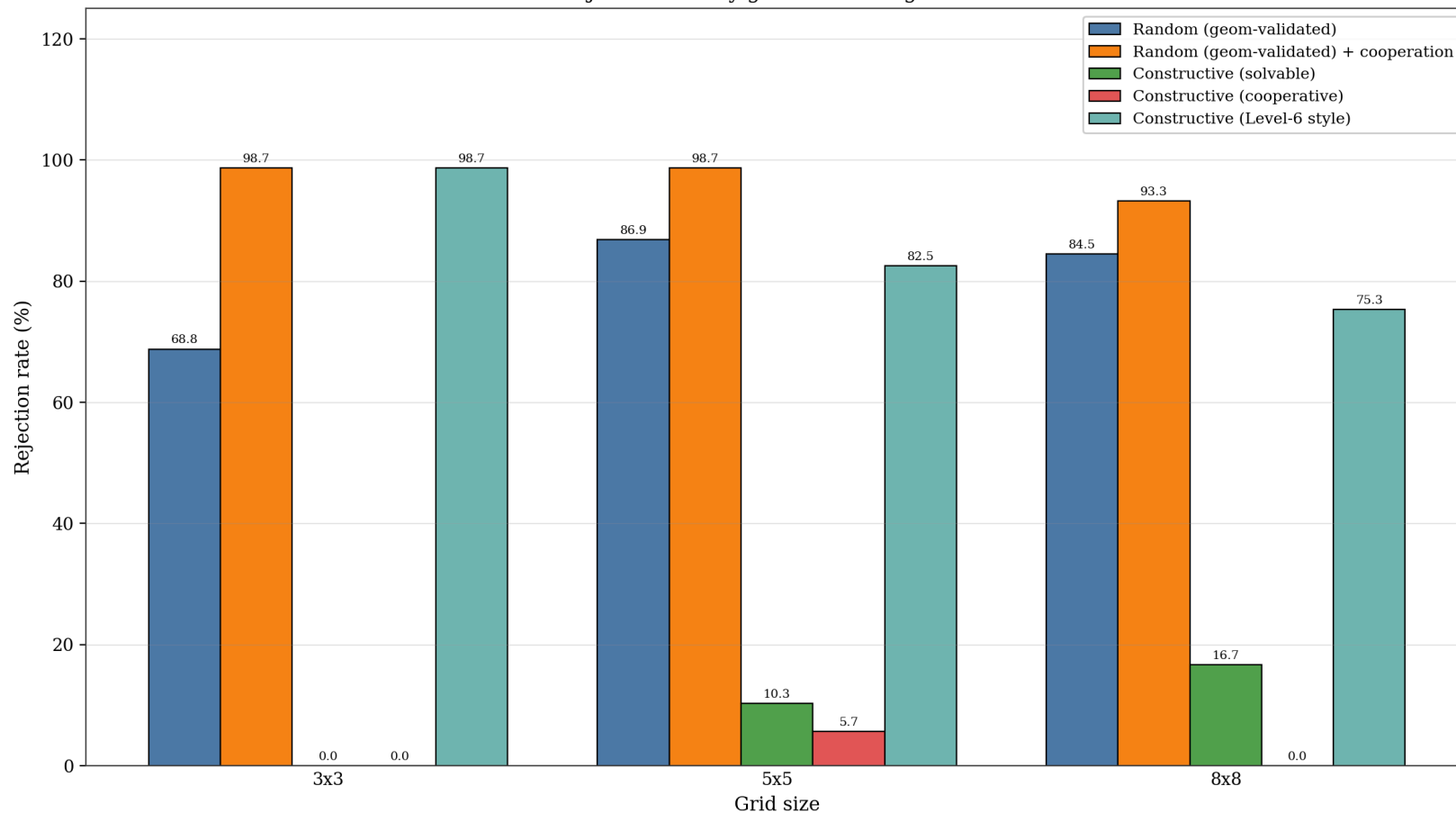


3. Add walls to shape the solution space



Generator efficiency (RQ3)

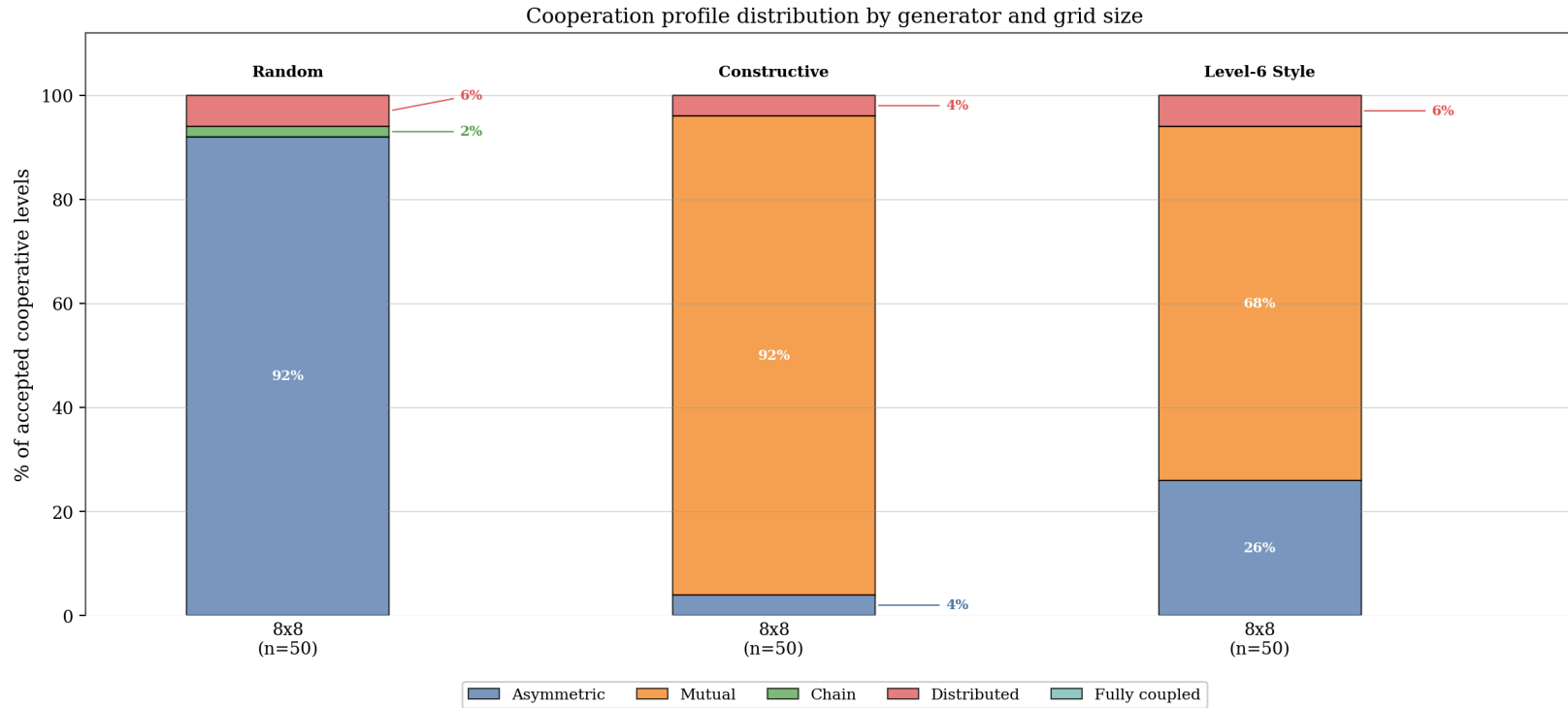
Rejection rate by generator and grid size



- 3x3 / 2-agents / 1-laser
- 5x5 / 3-agents / 2-lasers
- 8x8 / 4-agents / 3-lasers



Output diversity (RQ4)



- Constructive → 92% mutual.
- Level-6-style → 68% mutual.
- Random → mostly asymmetric.

(8x8 / 3-agents / 2-lasers)



Part 3 — Experiment (RQ5–6)

Can MARL agents trained on certified levels generalise to held-out ones, and does a staged curriculum outperform direct training on a hard target?

In this part we investigate whether generated levels are learnable:

RQ5 — Learnability

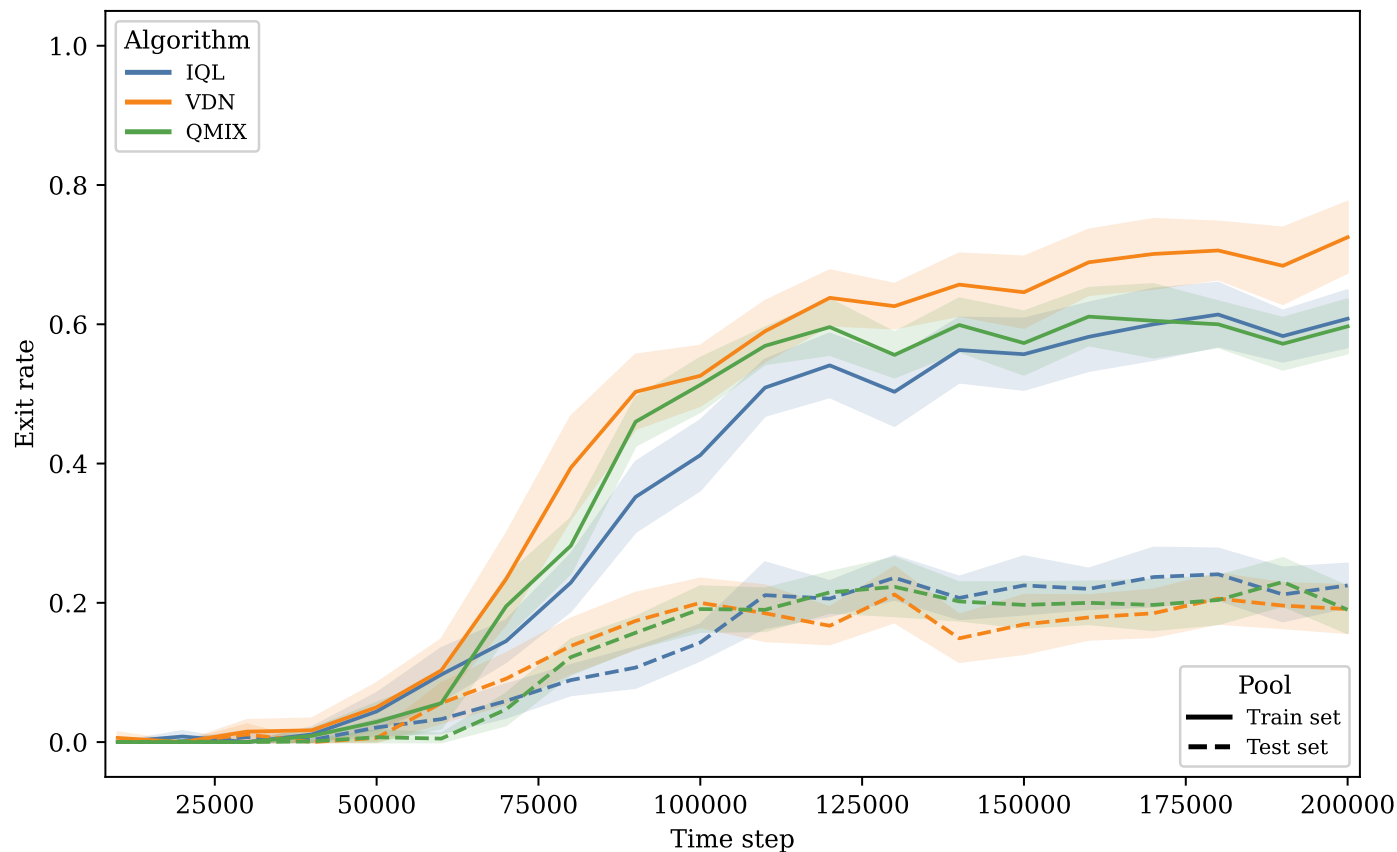
- Train IQL / VDN / QMIX on certified levels
- Measure train vs. test success
- Scale training data to close the generalisation gap

RQ6 — Curriculum learning

- Compare curriculum orderings (direct, forward, mixed, reverse)
- Test harder cooperation targets (mutual, Level-6)



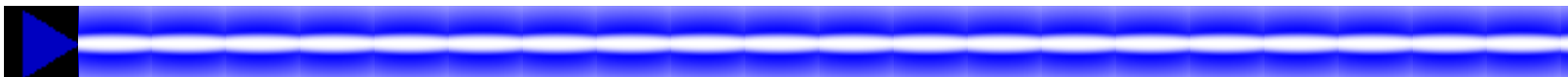
Learnability (RQ5)



(200k steps, 5x5 / 2-agents / 1-laser, asymmetric)

Takeaway

Learnable at 5×5, but too few levels $\Rightarrow$ overfitting



Scaling data (RQ5)



(300k steps, 5x5 / 2-agents / 1-laser, asymmetric)

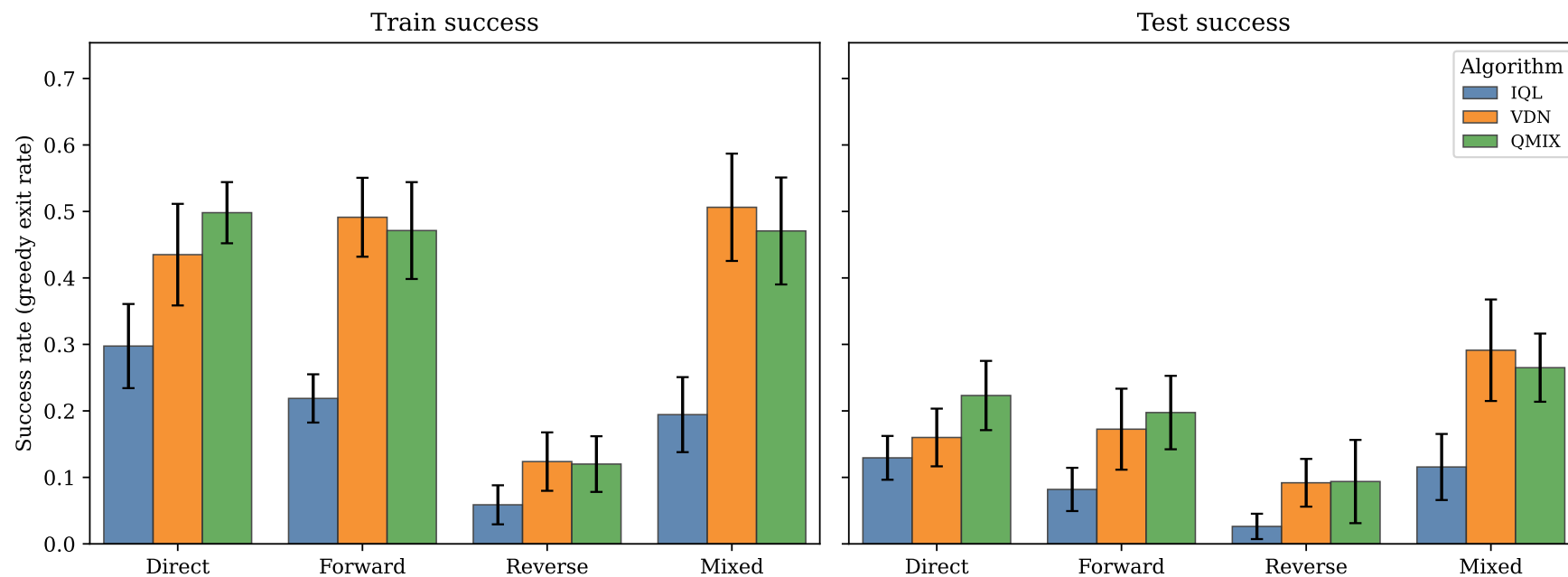
Takeaway

Unlimited certified levels $\Rightarrow$
overfitting becomes
generalisation.

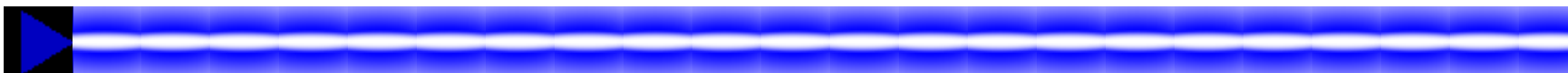


Curriculum ordering (RQ6)

Curriculum strategy on the 6x6/1L target (n=8 seeds, 95% CI)



Four budget-matched schedules (400k steps, 6×6 / 2-agents / 1-laser, asymmetric)



Curriculum targets (RQ6)

Target	Grid	Coop.	Budget	Test
Learnability	5×5	asym	200k	0.20
Curric. order	6×6	asym	400k	0.17
Frontier	6×6	mutual	600k	0.00
Frontier	5×5	mutual	200k	0.00
Level-6	12×13	mutual	2M	0.00

- Reachable target: ordering doesn't beat **direct**; only data **diversity** helps.
- A **cliff** at **mutual** cooperation: success → **zero**, even at 10× budget.

Takeaway

Scoped **negative** result: the base **mutual** task is unlearnable by IQL/VDN/QMIX.



Beyond the thesis: upstream

Solver, cooperation detector, and generators are **merged into the official LLE library** (`laser-learning-environment[generator]`, PyPI, v2.9.0+).

Public API

```
lle.solve • lle.is_cooperative • lle.cooperation_level • lle.generate
```

Takeaway

Not just thesis artefacts — **usable tooling** for the LLE users.



Conclusion

- RQ1 ✓ – SAT verification tool for solvability; in NP, $\leq_p$ SAT.
- RQ2 ✓ – strict counterfactual: $\text{coop.} \Leftrightarrow \text{SAT} \wedge \text{strict-UNSAT}$.
- RQ3 ✓ – SAT-as-oracle generators; every accepted level **certified**.
- RQ4 ✓ – five-profile taxonomy; generators steer toward **mutual**.
- RQ5 ✓ – certified levels learnable; scaling data closes the gap.
- RQ6 ✗ – value-based MARL can't learn **mutual** cooperation; no curriculum helps.

Takeaway

Central contribution: procedural generation coupled to **explicit certification**.



Future work

Theory & model

- Settle **NP-hardness** of LLE solvability.
- Add **gems** and true **void** tiles.
- Richer (time-indexed / weighted) cooperation graphs.

The certification framework is settled; the open question is **how far it extends**.

Learning

- **Make mutual coordination learnable:** dense / intrinsic rewards.
- Deeper research into cooperation structures: how do they affect learnability? Is the learning of one profile a stepping to another?
- Beyond value-based: actor-critics (MADDPG, COMA), MAVEN.



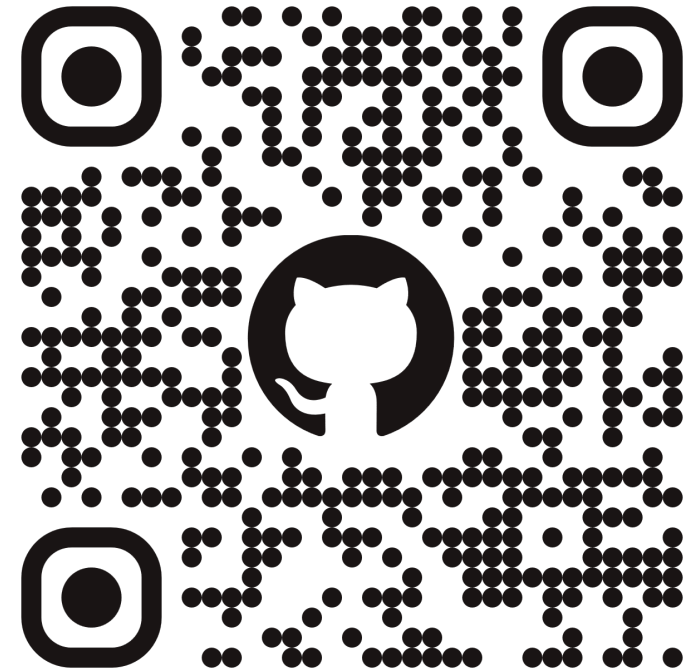
Summary

Contributions

1. SAT verification tool for LLE solvability.
2. Cooperation detector + five-profile taxonomy.
3. Solver-in-the-loop certified generators.
4. Merged into the official LLE library.

Key results

1. Constructive generator $\approx$ free.
2. Scaling certified data closes the generalisation gap.
3. A learnability **cliff** at mutual cooperation.



Backup slides